

Skill-7

1. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
ananya@DESKTOP-42I5F93:~$ mkdir -p skill7
ananya@DESKTOP-42I5F93:~$ cd skill7
ananya@DESKTOP-42I5F93:~/skill7$ pwd
/home/ananya/skill7
ananya@DESKTOP-42I5F93:~/skill7$ nano waitpid.c
ananya@DESKTOP-42I5F93:~/skill7$ ./waitpid
-bash: ./waitpid: No such file or directory
ananya@DESKTOP-42I5F93:~/skill7$ gcc waitpid.c -o waitpid
ananya@DESKTOP-42I5F93:~/skill7$ ./waitpid
Parent Process Started
Parent PID: 1056
```

Parent is waiting for Child...

```
Child Process Started
Child PID: 1057
Child is performing a task...
Child Process Completed
```

```
Child PID 1057 terminated normally
Child Exit Status: 0
Parent Process Resumed
Parent Process Completed
```

```
ananya@DESKTOP-42I5F93:~/skill7$ nano path_resolver.c
ananya@DESKTOP-42I5F93:~/skill7$ gcc path_resolver.c -o path_resolver
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver ls
Command found: /usr/bin/ls
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver gcc
Command found: /usr/bin/gcc
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver abcxyz
Command not found: abcxyz
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver pwd
Command found: /usr/bin/pwd
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver echo
Command found: /usr/bin/echo
ananya@DESKTOP-42I5F93:~/skill7$ ./path_resolver cat
Command found: /usr/bin/cat
ananya@DESKTOP-42I5F93:~/skill7$ ls
path_resolver  path_resolver.c  waitpid  waitpid.c
```

